

HDOC- II

Question:

Write a menu-driven C program using switch-case to input a positive integer and perform one of the following operations based on the user's choice:

- Find the sum of squares of its digits using a while loop.
- Find the sum of cubes of its digits using a for loop.
- Find the difference between the sum of cubes and the sum of squares of its digits using a do-while loop.

Task1: Algorithm:

1. Start.
2. Declare integer variables num, choice, temp, digit, sumSquares, sumCubes, and difference.
3. Initialize sumSquares = 0 and sumCubes = 0.
4. Input a positive integer num.
5. If num <= 0, display "Invalid input" and stop.
6. Display the menu:
 - 1 → Sum of squares of digits using a while loop
 - 2 → Sum of cubes of digits using a for loop
 - 3 → Difference between sum of cubes and sum of squares using a do-while loop
7. Input choice.
8. Use switch(choice) to perform the selected operation.
9. For Choice 1:
 - Set temp = num.
 - Repeat while temp > 0.
 - Extract the last digit using digit = temp % 10.
 - Calculate sumSquares = sumSquares + digit × digit.
 - Remove the last digit using temp = temp / 10.
 - Display sumSquares.
10. For Choice 2:
 - Use a for loop with temp = num.
 - Extract each digit.
 - Calculate sumCubes = sumCubes + digit × digit × digit.
 - Display sumCubes.

11. For Choice 3:

- Set temp = num.
- Using a do-while loop, extract each digit.
- Add its square to sumSquares.
- Add its cube to sumCubes.
- Remove the last digit.
- Calculate difference = sumCubes - sumSquares.
- Display difference.

12. For any other choice, display "Invalid choice".

13. Stop.

Task2: Test case Table :

Test Case	Number,Choice	Operation	Expected Output	Actual Output	Result
1	1234, 1	Sum of Squares	30	30	PASS
2	1234, 2	Sum of Cubes	100	100	PASS
3	1234,3	Difference	70	70	PASS
4	1234,4	Sum of Squares	Invalid choice	Invalid choice	PASS
5	456,3	Difference	328	328	PASS
6	456,2	Sum of cubes	405	405	PASS

Task3: C Program:

```
#include <stdio.h>
int main()
{
    int num, choice, temp, digit;
    int sumSquares = 0, sumCubes = 0, difference;
    printf("Enter a positive integer: ");
    scanf("%d", &num);

    if (num <= 0)
    {
        printf("Invalid input! Enter a positive integer.\n");
        return 0;
    }
    printf("\nMENU");
    printf("\n1. Sum of squares of digits");
    printf("\n2. Sum of cubes of digits");
    printf("\n3. Difference between sum of cubes and sum of squares");
    printf("\nEnter your choice: ");
    scanf("%d", &choice);

    switch(choice)
    {
        case 1:
            temp = num;
            while(temp > 0)
            {
                digit = temp % 10;
                sumSquares = sumSquares + (digit * digit);
                temp = temp / 10;
            }
            printf("\nSum of squares = %d\n", sumSquares);
            break;
        case 2:
            for(temp = num; temp > 0; temp = temp / 10)
            {
                digit = temp % 10;
                sumCubes = sumCubes + (digit * digit * digit);
            }
            printf("\nSum of cubes = %d\n", sumCubes);
            break;
        case 3:
            temp = num;
            do
            {
```

```

        digit = temp % 10;
        sumSquares = sumSquares + (digit * digit);
        sumCubes = sumCubes + (digit * digit * digit);

        temp = temp / 10;
    }
while(temp > 0);
    difference = sumCubes - sumSquares;

    printf("\nDifference = %d\n", difference);
    break;

default:
    printf("\nInvalid choice!\n");
}

return 0;
}

```

Task4: compilation:

```

(kali@kali)~/HDOC9.c/HDOC10.c/HDOC11.c
$ ./a.out
Enter a positive integer: 1234

1. Sum of squares of digits
2. Sum of cubes of digits
3. Difference between sum of cubes and sum of squares of digits
Enter your choice: 1

Sum of squares = 30

```

```

(kali@kali)~/HDOC9.c/HDOC10.c/HDOC11.c
$ ./a.out
Enter a positive integer: 1234

1. Sum of squares of digits
2. Sum of cubes of digits
3. Difference between sum of cubes and sum of squares of digits
Enter your choice: 2

Sum of cubes = 100

```

```
(kali㉿kali)-[~/HDOC9.c/HDOC10.c/HDOC11.c]
$ ./a.out
Enter a positive integer: 1234

1. Sum of squares of digits
2. Sum of cubes of digits
3. Difference between sum of cubes and sum of squares of digits
Enter your choice: 3

Difference = 70
```

```
(kali㉿kali)-[~/HDOC9.c/HDOC10.c/HDOC11.c]
$ ./a.out
Enter a positive integer: 1234

1. Sum of squares of digits
2. Sum of cubes of digits
3. Difference between sum of cubes and sum of squares of digits
Enter your choice: 4

Invalid choice!
```

```
(kali㉿kali)-[~/HDOC9.c/HDOC10.c/HDOC11.c]
$ ./a.out
Enter a positive integer: 456

1. Sum of squares of digits
2. Sum of cubes of digits
3. Difference between sum of cubes and sum of squares of digits
Enter your choice: 3

Difference = 328
```

```
(kali㉿kali)-[~/HDOC9.c/HDOC10.c/HDOC11.c]
$ ./a.out
Enter a positive integer: 456

1. Sum of squares of digits
2. Sum of cubes of digits
3. Difference between sum of cubes and sum of squares of digits
Enter your choice: 2

Sum of cubes = 405
```